

MongoDB Student Management Practical

1. Create/Select Database

use CollegeDB

Output:

switched to db CollegeDB

2. Create Students Collection

db.createCollection("students")

Output:

{ ok: 1 }

If you already created the collection, you can skip this step.

3. Insert 50 Students

This is the main dataset.

```
db.students.insertMany([
  {
    student_id "STU001",
    name: "Aarav Sharma",
    roll_no 101,
    age: 18,
    gender: "Male",
    course: "BCA",
    specialization: "Data Science & AI",
    marks: 82,
    city: "Delhi",
    address: "Delhi Main Road",
    father_name "Rajesh Sharma",
    phone: "9870000001"
  },
  {
    student_id "STU002",
    name: "Ananya Verma",
    roll_no 102,
    age: 19,
    gender: "Female",
    course: "BCA",
    specialization: "Data Science & AI",
    marks: 91,
    city: "Lucknow",
    address: "Lucknow Main Road",
    father_name "Suresh Verma",
    phone: "9870000002"
  },
  {
    student_id "STU003",
    name: "Aditya Singh",
```

```
roll_no 103,  
age: 20,  
gender: "Male",  
course: "BCA",  
specialization: "Data Science & AI",  
marks: 76,  
city: "Varanasi",  
address: "Varanasi Main Road",  
father_name "Mahesh Singh",  
phone: "9870000003"  
},  
{  
  student_id "STU004",  
  name: "Priya Gupta",  
  roll_no 104,  
  age: 18,  
  gender: "Female",  
  course: "BCA",  
  specialization: "Data Science & AI",  
  marks: 88,  
  city: "Kanpur",  
  address: "Kanpur Main Road",  
  father_name "Rakesh Gupta",  
  phone: "9870000004"  
},  
{  
  student_id "STU005",  
  name: "Arjun Kumar",  
  roll_no 105,  
  age: 19,  
  gender: "Male",  
  course: "BCA",  
  specialization: "Data Science & AI",  
  marks: 69,  
  city: "Patna",  
  address: "Patna Main Road",  
  father_name "Vijay Kumar",  
  phone: "9870000005"  
},  
{  
  student_id "STU006",  
  name: "Ishita Mishra",  
  roll_no 106,  
  age: 20,  
  gender: "Female",  
  course: "BCA",  
  specialization: "Data Science & AI",  
  marks: 95,
```

```
city: "Prayagraj",
address: "Prayagraj Main Road",
father_name "Anil Mishra",
phone: "9870000006"
},
{
  student_id "STU007",
  name: "Rohan Yadav",
  roll_no 107,
  age: 18,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 73,
  city: "Jaunpur",
  address: "Jaunpur Main Road",
  father_name "Dinesh Yadav",
  phone: "9870000007"
},
{
  student_id "STU008",
  name: "Sneha Tiwari",
  roll_no 108,
  age: 19,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 84,
  city: "Varanasi",
  address: "Varanasi Main Road",
  father_name "Manoj Tiwari",
  phone: "9870000008"
},
{
  student_id "STU009",
  name: "Karan Patel",
  roll_no 109,
  age: 20,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 61,
  city: "Ahmedabad",
  address: "Ahmedabad Main Road",
  father_name "Sanjay Patel",
  phone: "9870000009"
},
{
```

```
student_id "STU010",
name: "Neha Agarwal",
roll_no 110,
age: 18,
gender: "Female",
course: "BCA",
specialization: "Data Science & AI",
marks: 79,
city: "Agra",
address: "Agra Main Road",
father_name "Pankaj Agarwal",
phone: "9870000010"
},
{
student_id "STU011",
name: "Rahul Singh",
roll_no 111,
age: 19,
gender: "Male",
course: "BCA",
specialization: "Data Science & AI",
marks: 87,
city: "Gorakhpur",
address: "Gorakhpur Main Road",
father_name "Ashok Singh",
phone: "9870000011"
},
{
student_id "STU012",
name: "Simran Kaur",
roll_no 112,
age: 20,
gender: "Female",
course: "BCA",
specialization: "Data Science & AI",
marks: 93,
city: "Amritsar",
address: "Amritsar Main Road",
father_name "Harpreet Singh",
phone: "9870000012"
},
{
student_id "STU013",
name: "Vivek Pandey",
roll_no 113,
age: 18,
gender: "Male",
course: "BCA",
```

```
specialization: "Data Science & AI",
marks: 72,
city: "Ayodhya",
address: "Ayodhya Main Road",
father_name "Pramod Pandey",
phone: "9870000013"
},
{
  student_id "STU014",
  name: "Pooja Chauhan",
  roll_no 114,
  age: 19,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 89,
  city: "Dehradun",
  address: "Dehradun Main Road",
  father_name "Rajendra Chauhan",
  phone: "9870000014"
},
{
  student_id "STU015",
  name: "Yash Raj",
  roll_no 115,
  age: 20,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 66,
  city: "Ranchi",
  address: "Ranchi Main Road",
  father_name "Mohan Raj",
  phone: "9870000015"
},
{
  student_id "STU016",
  name: "Kavya Joshi",
  roll_no 116,
  age: 18,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 97,
  city: "Jaipur",
  address: "Jaipur Main Road",
  father_name "Deepak Joshi",
  phone: "9870000016"
```

```
},
{
  student_id "STU017",
  name: "Mohit Saxena",
  roll_no 117,
  age: 19,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 58,
  city: "Noida",
  address: "Noida Main Road",
  father_name "Ramesh Saxena",
  phone: "9870000017"
},
{
  student_id "STU018",
  name: "Riya Kapoor",
  roll_no 118,
  age: 20,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 81,
  city: "Delhi",
  address: "Delhi Main Road",
  father_name "Amit Kapoor",
  phone: "9870000018"
},
{
  student_id "STU019",
  name: "Aman Tripathi",
  roll_no 119,
  age: 18,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 74,
  city: "Lucknow",
  address: "Lucknow Main Road",
  father_name "Sunil Tripathi",
  phone: "9870000019"
},
{
  student_id "STU020",
  name: "Nandini Rai",
  roll_no 120,
  age: 19,
```

```
gender: "Female",
course: "BCA",
specialization: "Data Science & AI",
marks: 86,
city: "Varanasi",
address: "Varanasi Main Road",
father_name "Vinod Rai",
phone: "9870000020"
},
{
  student_id "STU021",
  name: "Dev Malhotra",
  roll_no 121,
  age: 20,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 92,
  city: "Chandigarh",
  address: "Chandigarh Main Road",
  father_name "Rajiv Malhotra",
  phone: "9870000021"
},
{
  student_id "STU022",
  name: "Sakshi Srivastava",
  roll_no 122,
  age: 18,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 77,
  city: "Lucknow",
  address: "Lucknow Main Road",
  father_name "Alok Srivastava",
  phone: "9870000022"
},
{
  student_id "STU023",
  name: "Harsh Vardhan",
  roll_no 123,
  age: 19,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 63,
  city: "Meerut",
  address: "Meerut Main Road",
```

```
father_name "Naresh Vardhan",
phone: "9870000023"
},
{
  student_id "STU024",
  name: "Muskan Khan",
  roll_no 124,
  age: 20,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 90,
  city: "Bhopal",
  address: "Bhopal Main Road",
  father_name "Salim Khan",
  phone: "9870000024"
},
{
  student_id "STU025",
  name: "Abhishek Dubey",
  roll_no 125,
  age: 18,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 71,
  city: "Prayagraj",
  address: "Prayagraj Main Road",
  father_name "Shyam Dubey",
  phone: "9870000025"
},
{
  student_id "STU026",
  name: "Tanya Mehta",
  roll_no 126,
  age: 19,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 85,
  city: "Surat",
  address: "Surat Main Road",
  father_name "Kishore Mehta",
  phone: "9870000026"
},
{
  student_id "STU027",
  name: "Ayush Tiwari",
```

```
roll_no 127,  
age: 20,  
gender: "Male",  
course: "BCA",  
specialization: "Data Science & AI",  
marks: 78,  
city: "Varanasi",  
address: "Varanasi Main Road",  
father_name "Sanjay Tiwari",  
phone: "9870000027"  
},  
{  
  student_id "STU028",  
  name: "Shreya Nair",  
  roll_no 128,  
  age: 18,  
  gender: "Female",  
  course: "BCA",  
  specialization: "Data Science & AI",  
  marks: 94,  
  city: "Kochi",  
  address: "Kochi Main Road",  
  father_name "Ramesh Nair",  
  phone: "9870000028"  
},  
{  
  student_id "STU029",  
  name: "Nikhil Jain",  
  roll_no 129,  
  age: 19,  
  gender: "Male",  
  course: "BCA",  
  specialization: "Data Science & AI",  
  marks: 68,  
  city: "Indore",  
  address: "Indore Main Road",  
  father_name "Mukesh Jain",  
  phone: "9870000029"  
},  
{  
  student_id "STU030",  
  name: "Aditi Bansal",  
  roll_no 130,  
  age: 20,  
  gender: "Female",  
  course: "BCA",  
  specialization: "Data Science & AI",  
  marks: 83,
```

```
city: "Delhi",
address: "Delhi Main Road",
father_name "Rohit Bansal",
phone: "9870000030"
},
{
  student_id "STU031",
  name: "Varun Mishra",
  roll_no 131,
  age: 18,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 57,
  city: "Kanpur",
  address: "Kanpur Main Road",
  father_name "Ravi Mishra",
  phone: "9870000031"
},
{
  student_id "STU032",
  name: "Diya Chawla",
  roll_no 132,
  age: 19,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 96,
  city: "Chandigarh",
  address: "Chandigarh Main Road",
  father_name "Vikas Chawla",
  phone: "9870000032"
},
{
  student_id "STU033",
  name: "Manish Thakur",
  roll_no 133,
  age: 20,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 75,
  city: "Shimla",
  address: "Shimla Main Road",
  father_name "Devendra Thakur",
  phone: "9870000033"
},
{
```

```
student_id "STU034",
name: "Komal Soni",
roll_no 134,
age: 18,
gender: "Female",
course: "BCA",
specialization: "Data Science & AI",
marks: 80,
city: "Jaipur",
address: "Jaipur Main Road",
father_name "Rajesh Soni",
phone: "9870000034"
},
{
student_id "STU035",
name: "Siddharth Roy",
roll_no 135,
age: 19,
gender: "Male",
course: "BCA",
specialization: "Data Science & AI",
marks: 64,
city: "Kolkata",
address: "Kolkata Main Road",
father_name "Subhash Roy",
phone: "9870000035"
},
{
student_id "STU036",
name: "Meera Iyer",
roll_no 136,
age: 20,
gender: "Female",
course: "BCA",
specialization: "Data Science & AI",
marks: 87,
city: "Bengaluru",
address: "Bengaluru Main Road",
father_name "Krishna Iyer",
phone: "9870000036"
},
{
student_id "STU037",
name: "Akash Srivastava",
roll_no 137,
age: 18,
gender: "Male",
course: "BCA",
```

```
specialization: "Data Science & AI",
marks: 70,
city: "Ayodhya",
address: "Ayodhya Main Road",
father_name "Rakesh Srivastava",
phone: "9870000037"
},
{
  student_id "STU038",
  name: "Pallavi Sharma",
  roll_no 138,
  age: 19,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 91,
  city: "Delhi",
  address: "Delhi Main Road",
  father_name "Vivek Sharma",
  phone: "9870000038"
},
{
  student_id "STU039",
  name: "Gaurav Mehta",
  roll_no 139,
  age: 20,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 59,
  city: "Surat",
  address: "Surat Main Road",
  father_name "Nitin Mehta",
  phone: "9870000039"
},
{
  student_id "STU040",
  name: "Tanvi Gupta",
  roll_no 140,
  age: 18,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 88,
  city: "Agra",
  address: "Agra Main Road",
  father_name "Amit Gupta",
  phone: "9870000040"
```

```
},
{
  student_id "STU041",
  name: "Ritik Yadav",
  roll_no 141,
  age: 19,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 73,
  city: "Jaunpur",
  address: "Jaunpur Main Road",
  father_name "Mahendra Yadav",
  phone: "9870000041"
},
{
  student_id "STU042",
  name: "Bhavna Joshi",
  roll_no 142,
  age: 20,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 82,
  city: "Dehradun",
  address: "Dehradun Main Road",
  father_name "Suresh Joshi",
  phone: "9870000042"
},
{
  student_id "STU043",
  name: "Abhinav Kumar",
  roll_no 143,
  age: 18,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 99,
  city: "Patna",
  address: "Patna Main Road",
  father_name "Raj Kumar",
  phone: "9870000043"
},
{
  student_id "STU044",
  name: "Manya Singh",
  roll_no 144,
  age: 19,
```

```
gender: "Female",
course: "BCA",
specialization: "Data Science & AI",
marks: 76,
city: "Gorakhpur",
address: "Gorakhpur Main Road",
father_name "Vijay Singh",
phone: "9870000044"
},
{
  student_id "STU045",
  name: "Saurabh Verma",
  roll_no 145,
  age: 20,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 67,
  city: "Lucknow",
  address: "Lucknow Main Road",
  father_name "Ashish Verma",
  phone: "9870000045"
},
{
  student_id "STU046",
  name: "Isha Kapoor",
  roll_no 146,
  age: 18,
  gender: "Female",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 89,
  city: "Delhi",
  address: "Delhi Main Road",
  father_name "Manoj Kapoor",
  phone: "9870000046"
},
{
  student_id "STU047",
  name: "Ravi Shukla",
  roll_no 147,
  age: 19,
  gender: "Male",
  course: "BCA",
  specialization: "Data Science & AI",
  marks: 62,
  city: "Varanasi",
  address: "Varanasi Main Road",
```

```
    father_name "Dinesh Shukla",
    phone: "9870000047"
  },
  {
    student_id "STU048",
    name: "J hanvi Patel",
    roll_no 148,
    age: 20,
    gender: "Female",
    course: "BCA",
    specialization: "Data Science & AI",
    marks: 93,
    city: "Ahmedabad",
    address: "Ahmedabad Main Road",
    father_name "Rakesh Patel",
    phone: "9870000048"
  },
  {
    student_id "STU049",
    name: "Tarun Agarwal",
    roll_no 149,
    age: 18,
    gender: "Male",
    course: "BCA",
    specialization: "Data Science & AI",
    marks: 71,
    city: "Agra",
    address: "Agra Main Road",
    father_name "Suresh Agarwal",
    phone: "9870000049"
  },
  {
    student_id "STU050",
    name: "Priti Nair",
    roll_no 150,
    age: 19,
    gender: "Female",
    course: "BCA",
    specialization: "Data Science & AI",
    marks: 84,
    city: "Kochi",
    address: "Kochi Main Road",
    father_name "Mohan Nair",
    phone: "9870000050"
  }
}
```

)

Output:

acknowledged: true

```
insertedIds { ...50 IDs... }
```

4. Display All Students

```
db.students.find()
```

Output

It displays all 50 student documents

5. Display Data in Readable Format

```
db.students.find().pretty()
```

This is useful for taking screenshots for your practical file.

6. Display Only Name and Marks

This is called **Projection**

```
db.students.find(
```

```
{},
{
  name: 1,
  marks: 1,
  _id: 0
})
```

Example output

```
{ name "Aarav Sharma", mark82 }
{ name "Ananya Verma", mark91 }
{ name "Aditya Singh", mark76 }
```

...

7. Students Having Marks Greater Than 80

```
db.students.find(
  { marks { $gt: 80 } },
  { name 1, marks: 1, _id0 }
)
```

Meaning: \$gt = greater than.

Example:

```
{ name "Aarav Sharma", mark82 }
{ name "Ananya Verma", mark91 }
{ name "Priya Gupta", mark88 }
```

...

8. Students Having Marks Greater Than or Equal to 80

```
db.students.find(
  { marks { $gte: 80 } },
  { name 1, marks: 1, _id0 }
)
```

\$gte = greater than or equal to.

9. Students Having Marks Less Than 70

```
db.students.find(
  { marks { $lt: 70 } },
  { name 1, marks: 1, _id: 0 }
)
```

\$lt = less than.

Example:

Arjun Kumar - 69

Karan Patel - 61

Yash Raj - 66

Mohit Saxena - 58

...

10. Students From Varanasi

```
db.students.find(
  { city: "Varanasi" },
  { name 1, city: 1, _id: 0 }
)
```

Output

```
{ name "Aditya Singh", city: "Varanasi" }
{ name "Sneha Tiwari", city: "Varanasi" }
{ name "Nandini Rai", city: "Varanasi" }
{ name "Ayush Tiwari", city: "Varanasi" }
{ name "Ravi Shukla", city: "Varanasi" }
```

11. Students From Delhi OR Lucknow

```
db.students.find(
  {
    $or: [
      { city: "Delhi" },
      { city: "Lucknow" }
    ]
  },
  {
    name: 1,
    city: 1,
    _id: 0
  }
)
```

\$or means **either condition can be true**

12. Marks Between 70 and 90

```
db.students.find(
  {
    marks: {
      $gte: 70,
      $lte: 90
    }
  },
)
```

```
{
  name: 1,
  marks: 1,
  _id: 0
}
)
$gte = greater than/equal
$lte = less than/equal
```

13. Students From Selected Cities Using \$in

```
db.students.find(
{
  city: {
    $in: ["Delhi", "Varanasi", "Lucknow"]
  }
},
{
  name: 1,
  city: 1,
  _id: 0
}
)
$in checks whether a value belongs to a list.
```

14. Students NOT From Delhi

```
db.students.find(
{
  city: {
    $ne: "Delhi"
  }
},
{
  name: 1,
  city: 1,
  _id: 0
}
)
$ne = not equal.
```

15. Count Total Students

```
db.students.countDocuments()
```

Output

50

16. Count Students With Marks ≥ 80

```
db.students.countDocuments({
  marks: { $gte: 80 }
})
```

Output

25

17. Display Different Cities

```
db.studentsdistinct("city")
```

Output

```
[
  "Delhi",
  "Lucknow",
  "Varanasi",
  "Kanpur",
  "Patna",
  "Prayagraj",
  "Jaunpur",
  "Ahmedabad",
  "Agra",
  "Gorakhpur",
  "Amritsar",
  "Ayodhya",
  "Dehradun",
  "Ranchi",
  "Jaipur",
  "Noida",
  "Chandigarh",
  "Meerut",
  "Bhopal",
  "Surat",
  "Kochi",
  "Indore",
  "Shimla",
  "Kolkata",
  "Bengaluru"
]
```

18. Sort Students by Marks — Highest First

```
db.students.find(
```

```
  {},
  {
    name: 1,
    marks: 1,
    _id: 0
  }
)
```

```
  ).sort({
    marks: -1
  })
```

-1 = descending order.

19. Top 5 Students

```

db.students.find(
  {},
  {
    name: 1,
    marks: 1,
    _id: 0
  }
).sort({
  marks: -1
}).limit(5)

```

Output

```

{ name "Abhinav Kumar", marks 89 }
{ name "Kavya Joshi", marks 97 }
{ name "Diya Chawla", marks 96 }
{ name "Ishita Mishra", marks 95 }
{ name "Shreya Nair", marks 94 }

```

20. Skip First 5 Students

```

db.students.find(
  {},
  {
    name: 1,
    marks: 1,
    _id: 0
  }
).sort({
  roll_no: 1
}).skip(5).limit(5)

```

21. Update One Student

Change Aarav's marks from 82 to 85.

```

db.students.updateOne(
  { student_id: "STU001" },
  {
    $set: {
      marks: 85
    }
  }
)

```

Output

```

matchedCount 1
modifiedCount 1

```

22. Update Many Students

Give students with marks ≥ 90 an "Excellent" performance label.

```

db.students.updateMany(
  {

```

```

    marks:{ $gte: 90 }
  },
  {
    $set: {
      performance: "Excellent"
    }
  }
)

```

Output

matchedCount 9
modifiedCount 9

23. Increase Marks Using \$ inc

```

db.students.updateOne(
  { student_id: "STU005" },
  {
    $inc: {
      marks: 2
    }
  }
)

```

Arjun's marks become:
69 → 71

24. Add a New Field

```

db.students.updateOne(
  { student_id: "STU010" },
  {
    $set: {
      scholarship: true
    }
  }
)

```

25. Remove a Field Using \$unset

```

db.students.updateOne(
  { student_id: "STU010" },
  {
    $unset: {
      scholarship: ""
    }
  }
)

```

26. Delete One Student

 Do this only when you need to demonstrate deletion.

```

db.students.deleteOne(
  { student_id: "STU050" }
)

```

```
}}
```

Output

deletedCount 1

Tip: Perform this near the end so that your 50-student dataset remains intact for the other practicals.

27. Delete Multiple Students

For example, delete students whose marks are below 60.

```
db.students.deleteMany({
  marks: {
    $lt: 60
  }
})
```

Output

deletedCount 2

Again, do this **at the end**

28. Find Average Marks

This is an **Aggregation** operation.

```
db.students.aggregate([
  {
    $group: {
      _id: null,
      average_marks: {
        $avg: "$marks"
      }
    }
  }
])
```

Output

```
{ average_marks 78.82 }
```

29. Find Highest Marks

```
db.students.aggregate([
  {
    $group: {
      _id: null,
      highest_marks: {
        $max: "$marks"
      }
    }
  }
])
```

Output

```
{ highest_marks 99 }
```

30. Find Lowest Marks

```
db.students.aggregate([
  {
    $group: {
      _id: null,
      lowest_marks: {
        $min: "$marks"
      }
    }
  }
])
```

Output

```
{ lowest_marks: 57 }
```

31. Find Total Marks

```
db.students.aggregate([
  {
    $group: {
      _id: null,
      total_marks: {
        $sum: "$marks"
      }
    }
  }
])
```

Output

```
{ total_marks: 3941 }
```

32. City-Wise Average Marks

```
db.students.aggregate([
  {
    $group: {
      _id: "$city",
      average_marks: {
        $avg: "$marks"
      },
      student_count: {
        $sum: 1
      }
    }
  },
  {
    $sort: {
      average_marks: -1
    }
  }
])
```

```
)
```

Example output

```
{_id: "Amritsar", average_marks: 93, student_count: 1 }
{ _id: "Kochi", average_marks: 89, student_count: 2 }
{ _id: "Chandigarh", average_marks: 94, student_count: 2 }
```

...

33. Count Students City-Wise

```
db.students.aggregate([
```

```
{
  $group: {
    _id: "$city",
    total_students: {
      $sum: 1
    }
  }
},
```

```
{
  $sort: {
    total_students: -1
  }
}]
```

```
)
```

Example

```
{_id: "Delhi", total_students: 5 }
{ _id: "Varanasi", total_students: 5 }
{ _id: "Lucknow", total_students: 4 }
```

...

34. \$match + \$sort + \$project

Find students with marks ≥ 80 and display them from highest to lowest.

```
db.students.aggregate([
```

```
{
  $match: {
    marks: {
      $gte: 80
    }
  }
},
```

```
{
  $sort: {
    marks: -1
  }
},
```

```
{
  $project: {
    _id: 0,
```

```

    name: 1,
    marks: 1
  }
}
})

```

35. Create Index

Create an index on roll number:

```

db.students.createIndex({
  roll_no: 1
})

```

Output

```

roll_no_1

```

36. Display Indexes

```

db.students.getIndexes()

```

Output

You will see something similar to:

```

[
  {
    name: "_id_",
    key: { _id: 1 }
  },
  {
    name: "roll_no_1",
    key: { roll_no: 1 }
  }
]

```

37. Regular Expression Search

Find students whose names start with A.

```

db.students.find(
  {
    name: /^A/
  },
  {
    name: 1,
    _id: 0
  }
)

```

Output

```

{ name: "Aarav Sharma" }
{ name: "Ananya Verma" }
{ name: "Aditya Singh" }
{ name: "Arjun Kumar" }
{ name: "Aman Tripathi" }
{ name: "Aditi Bansal" }

```

```
{ name "Ayush Tiwari"}  
{ name "Abhishek Dubey"}  
{ name "Akash Srivastava"}  
{ name "Abhinav Kumar"}
```

38. Explain Query

```
db.students.find(  
  roll_no:127  
)>.explain("executionStats")
```

This shows how MongoDB executes the query and provides information about the index/query execution.